# GEOGRAPHY

# PHYSICAL GEOGRAPHY

# GEOMORPHOLOGY

## MODULE - 17

### VOLCANOES

- WHAT ARE VOLCANOES?
- CLASSIFICATION OF VOLCANOES

# GEOMORPHOLOGY

## PLAN OF ACTION

### WHAT GVR SIR HAS COVERED

- EARTH'S INTERIOR
- SOURCES OF HEAT IN EARTH'S INTERIOR
- CONTINENTAL DRIFT THEORY
- PLATE TECTONICS THEORY
- CONVERGENT, DIVERGENT & TRANSFORM PLATE BOUNDARIES

### WHAT I WILL COVER

- VOLCANOES
- EARTHQUAKES
- GEOMORPHIC PROCESSES
- LANDFORMS & THEIR EVOLUTION
  1. FLUVIAL LANDFORMS
  2. GLACIAL LANDFORMS
  3. ARID LANDFORMS
  4. KARST TOPOGRAPHY
  5. COASTAL LANDFORMS

# GEOMORPHOLOGY

## PLAN OF ACTION

### WHAT WILL HAPPEN TO CLIMATOLOGY?

*It will continue as usual…*

| | |
|---|---|
| **Atmosphere** | ▪ Composition and structure; weather and climate; |
| **Insolation** | ▪ Factors controlling insolation; heat budget of the earth – heating and cooling of atmosphere (conduction, convection, terrestrial radiation, advection); |
| **Temperature** | ▪ Factors controlling temperature; distribution of temperature – horizontal and vertical; inversion of temperature; |
| **Pressure** | ▪ Pressure belts; winds-planetary, seasonal and local; air masses and fronts; tropical and extra-tropical cyclones; |
| **Precipitation** | ▪ Evaporation; condensation – dew, frost, fog, mist and cloud; rainfall – types and world distribution; |
| **World Climate** | ▪ Different types of Climate and Climate changes. |

# VOLCANOES

## WHAT IS A VOLCANO?


1 The ancient Greeks believed that volcanic eruptions occurred when Vulcan, the God of the Underworld, stoked his subterranean furnace beneath Vulcano, a small volcanic island off Sicily, from which the present word volcano is derived.

***It is different story, we no longer believe this is true.***

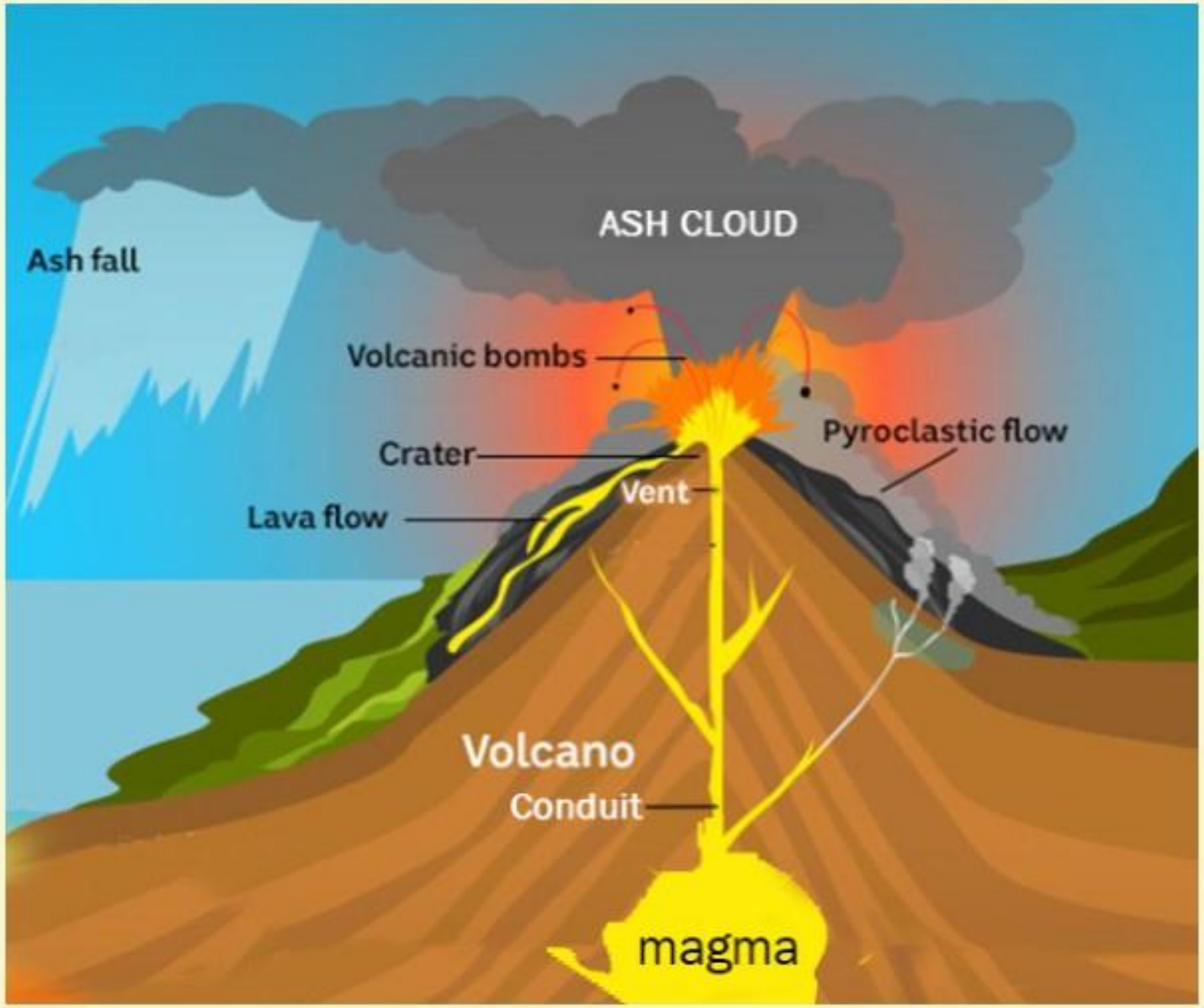
# VOLCANOES

## VOLCANO AND VULCANISM

**2** A volcano is a vent or opening nearly circular in form through which heated materials consisting of gases, liquid lava and fragments of rocks are ejected from the highly heated interior to the earth's surface.

**3** All phenomena connected with the movement heated material from the interior towards the surface of the earth is known as vulcanism.

# VOLCANOES

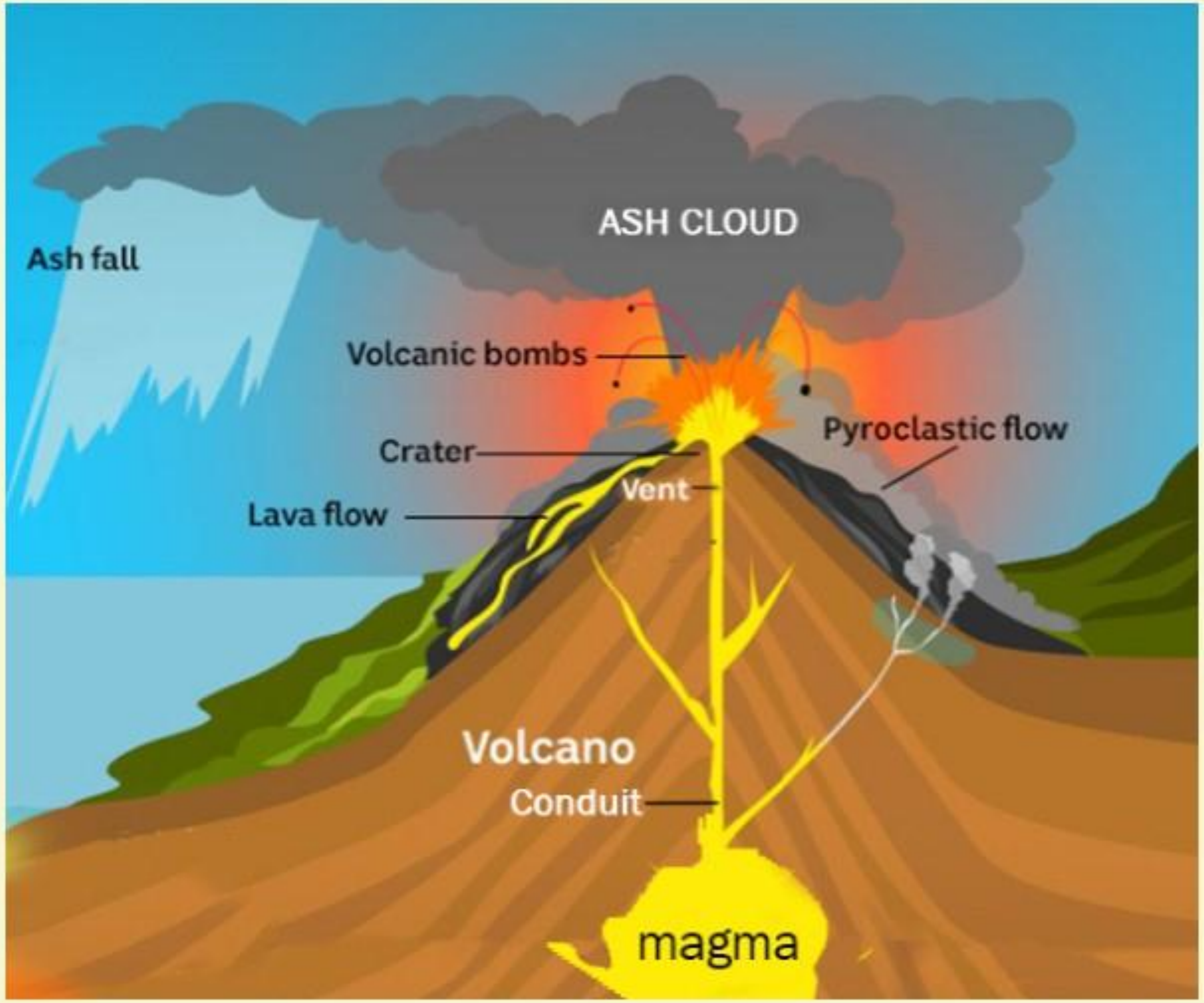



## WHAT ARE THE VARIOUS COMPONENTS OF VOLCANO?

1. Magma - Molten rock beneath Earth's surface.
2. Vent - An opening in Earth's surface through which volcanic materials escape.
3. Lava - Molten rock that erupts from a volcano and which solidifies as it cools.
4. Crater - Mouth of a volcano that surrounds a volcanic vent.
5. Conduit - An underground passage through which magma travels.

# VOLCANOES






## WHAT ARE THE VARIOUS COMPONENTS OF VOLCANO?

6. Ash - Fragments of lava or rock smaller than 2 mm in size that are blasted into the air by volcanic explosions.
7. Ash Cloud - A cloud of ash formed by volcanic explosions.
8. Pyroclasts - The coarser fragmental rocks that include lapilli, pumice and volcanic bombs.
9. The finer particles are the Volcanic Dust, which may be shot so high into the sky that it travels round the world several times before it eventually comes to rest.

# VOLCANOES

## WHAT ARE VOLCANOES?

### MAGMA vs LAVA

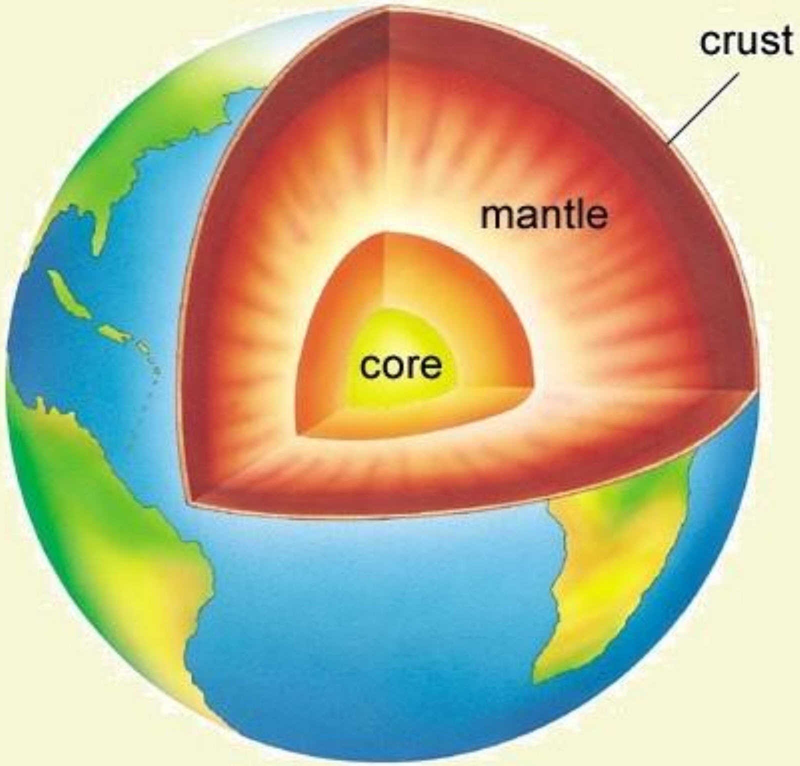
1. Magma is a hot fluid material below or within the earth's crust from which lava and other igneous rocks are formed.

2. When magma is ejected by a volcano or other vent, the material is called lava.

3. Magma that has cooled into a solid is called igneous rock.

# VOLCANOES

## WHAT ARE VOLCANOES?




### MAGMA vs LAVA

1. A weaker zone of the mantle called asthenosphere, usually is the source of magma.

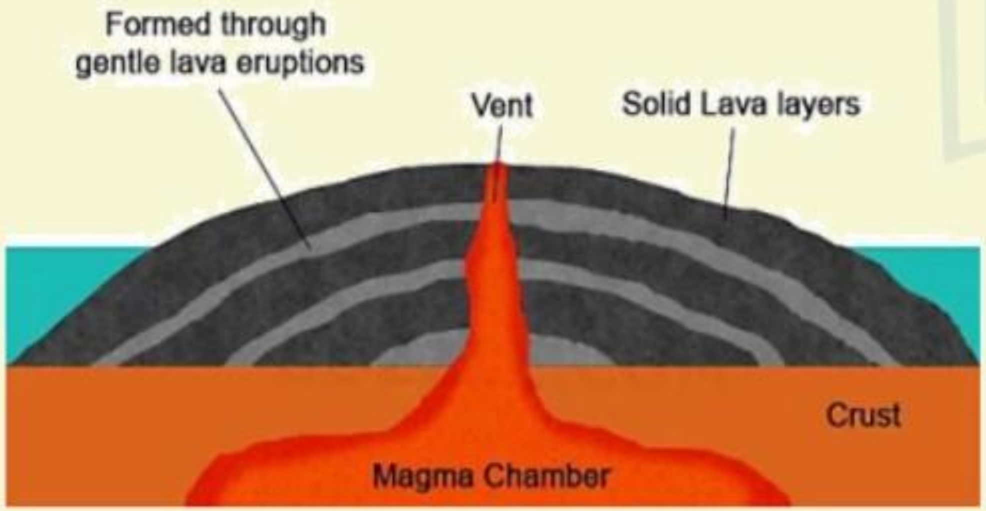
2. Once this magma comes out to the earth surface through the vent of a volcano, it is called as the Lava. Therefore, Lava is nothing but the magma on earth surface.

3. The process by which solid, liquid and gaseous material escape from the earth's interior to the surface of the earth is called as Volcanism.

# VOLCANOES

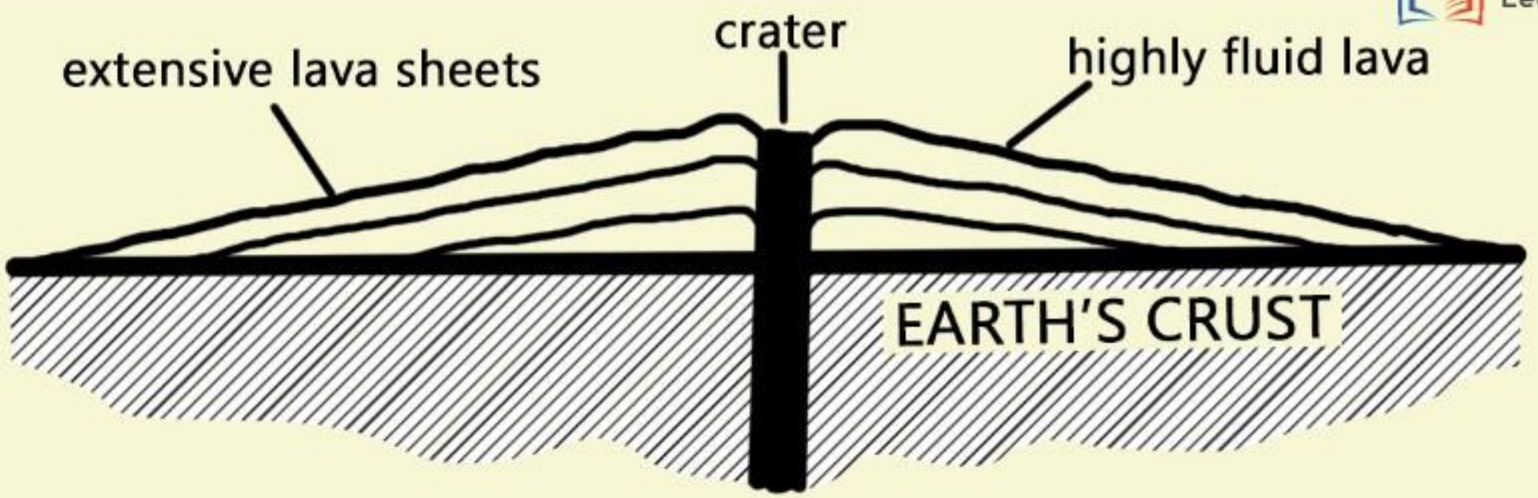
## BASIC LAVAS

The Anatomy of a Shield Volcano

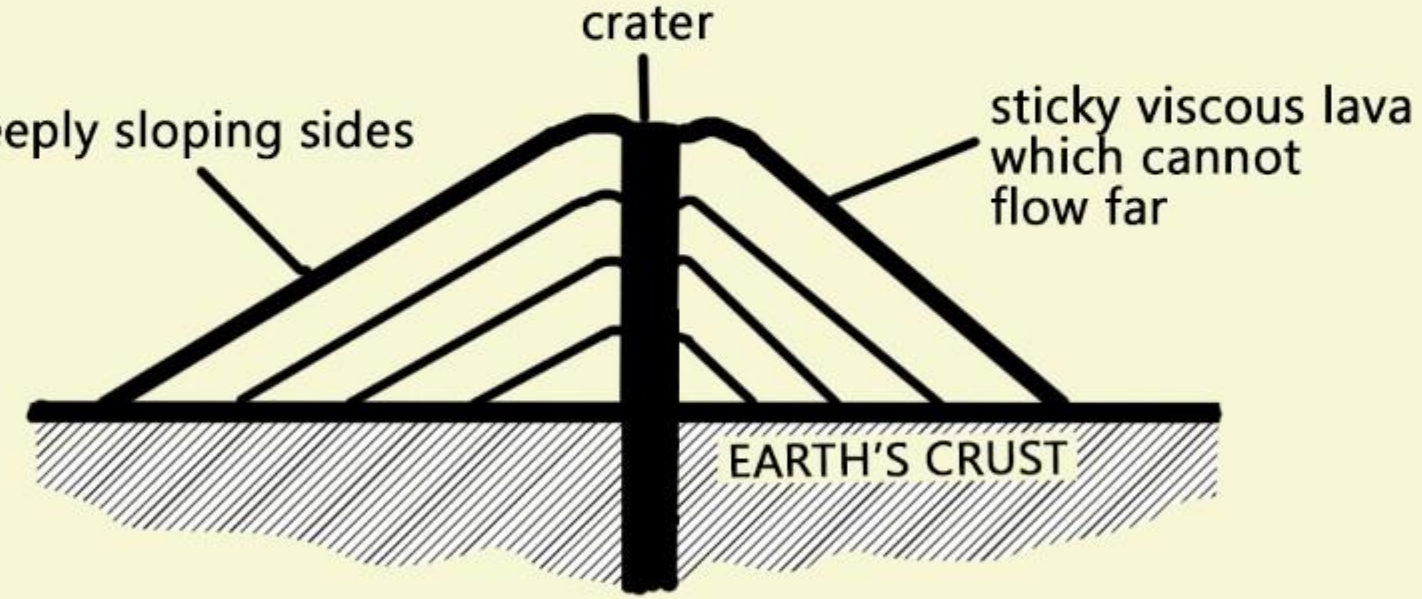



1. These are the hottest lavas, about 1,000ºC and are highly fluid.
2. They are dark coloured like basalt, rich in iron and magnesium but poor in silica.
3. Once they come out of the vent, they flow quietly and they are not very explosive.
4. They flow readily with a speed of 10 to 30 miles per hour.
5. They affect extensive areas, spreading out as thin sheets over great distances before they solidify.
6. The resultant volcano is gently sloping with a wide diameter and forms a flattened shield or dome.

## LAVA DOME OR SHIELD VOLCANO



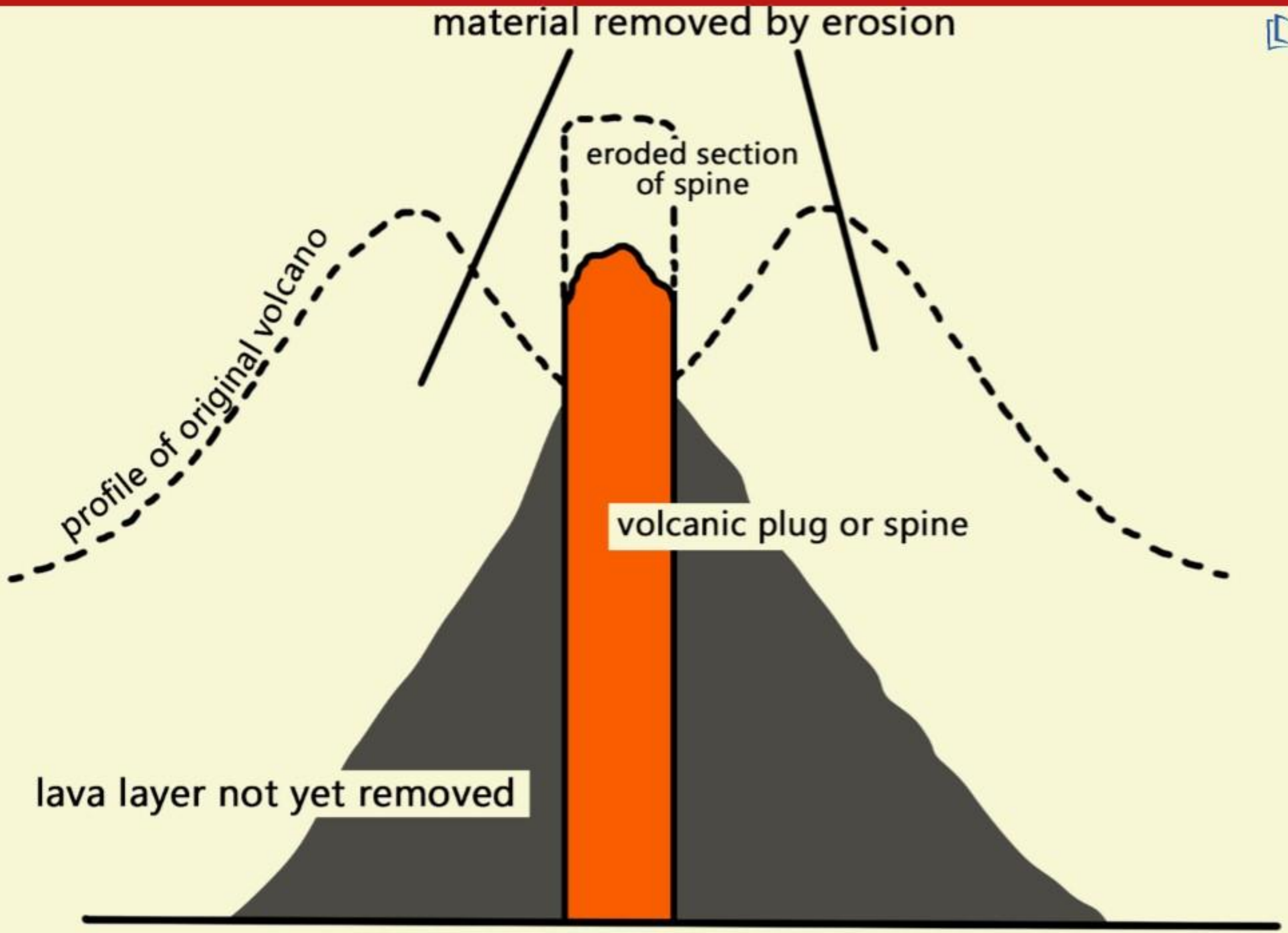

## ACID LAVA CONE



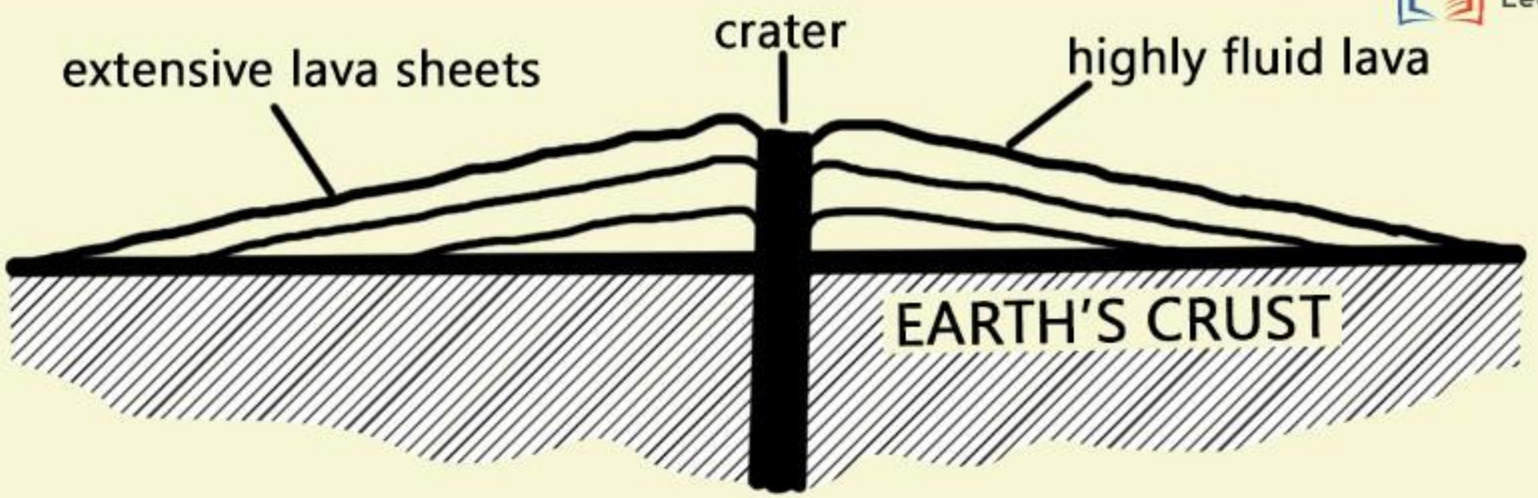
# VOLCANOES

## ACID LAVAS

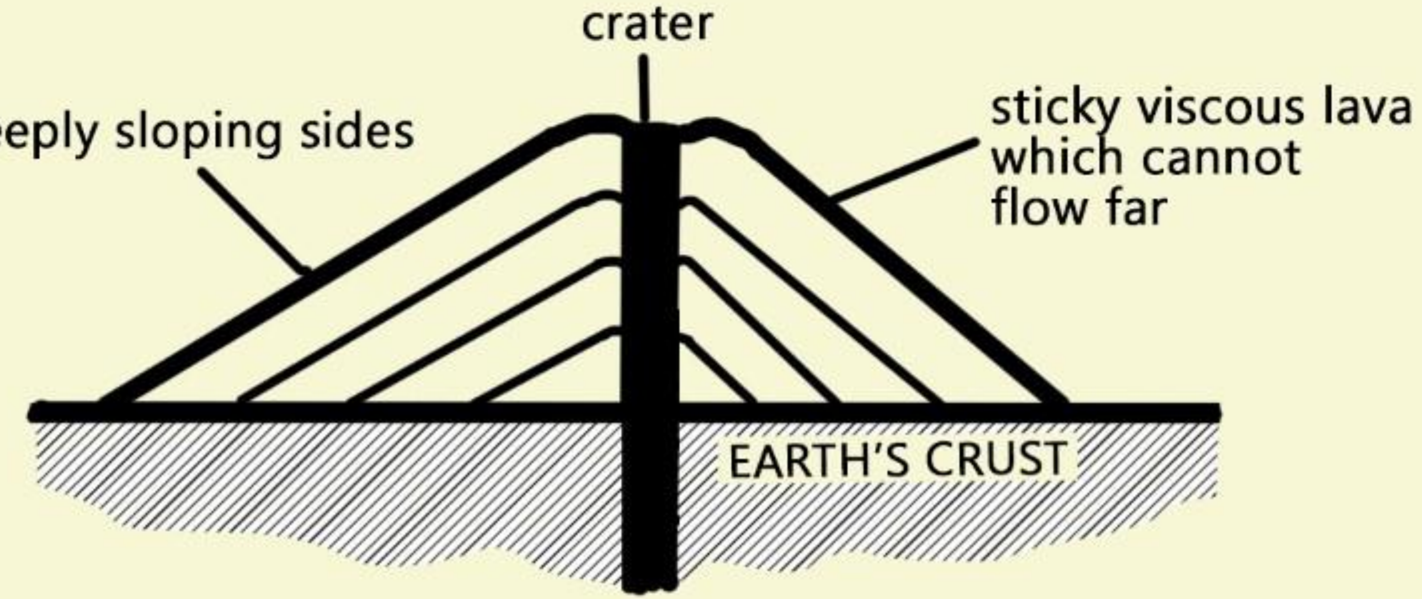
1. These lavas are highly viscous with high melting point.
2. They are light coloured, low density, high percentage of silica.
3. They flow slowly and seldom travel far before solidifying.
4. The resultant cone is steep-sided.
5. The rapid solidification of the lava in the vent obstruct the flow of the out-pouring lava that results in loud explosions, throwing out many volcanic bombs or pyroclasts.
6. Sometimes the lavas are so viscous that they form a spine or plug at the crater like that of Mount Pelee in Martinique.


material removed by erosion
eroded section
of spine
profile of original volcano
volcanic plug or spine
lava layer not yet removed

LAVA DOME OR SHIELD VOLCANO




ACID LAVA CONE



# VOLCANOES

Mount Pelee In Martinique

## NEEDLE OF PELEE

Beginning in October 1902, a dramatic volcanic spine grew from the crater floor reaching a height of about 300m. Called the "Needle of Pelée" or "Pelée's Tower", this extraordinary volcanic feature rose up to 15 m (50 ft) a day, and became twice the height of the Washington Monument. It became unstable and collapsed into a pile of rubble in March 1903 after 5 months of growth.

# CLASSIFICATION OF VOLCANOES

## CLASSIFICATION ON BASIS OF MODE OF ERUPTION

### CENTRAL ERUPTION TYPE OR EXPLOSIVE ERUPTION TYPE

- Hawaiian Type
- Strombolian Type
- Vulcanian Type
- Pelean Type
- Vesuvius Type

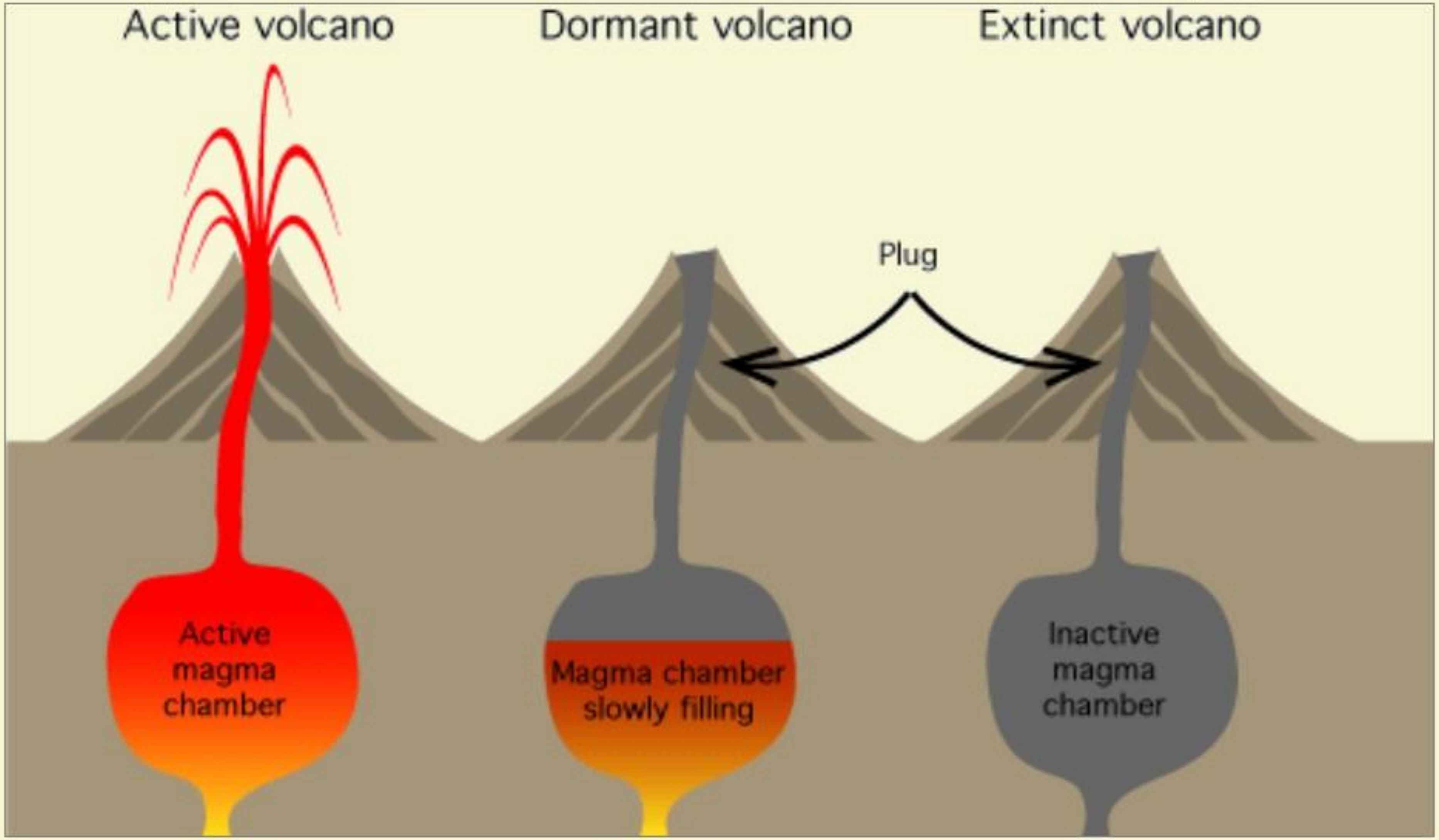
### FISSURE ERUPTION TYPE

## CLASSIFICATION ON THE BASIS OF PERIODICITY OF ERUPTION

- Active Volcanoes
- Dormant Volcanoes
- Extinct Volcanoes

# CLASSIFICATION OF VOLCANOES

## 1 CLASSIFICATION ON THE BASIS OF PERIODICITY OF ERUPTION



# CLASSIFICATION OF VOLCANOES

## 1 CLASSIFICATION ON THE BASIS OF PERIODICITY OF ERUPTION


Mt Etna in Sicily

### ACTIVE VOLCANOES

1. Active Volcanoes are those which constantly eject volcanic lavas, gases, ashes and fragmental materials.
2. Most of the active volcanoes are found along the mid oceanic ridges representing divergent plate margins (constructive plate margins) and in convergent plate margins (destructive plate margins).
3. Mount Etna and Mount Stromboli in the Mediterranean sea are the most significant examples of this category.


GERMANY
BELGIUM
UKRAINE
CZECH REP
SLOVAKIA
MOLDOVA
AUSTRIA
HUNGARY
FRANCE
SWITZERLAND
SLOVENIA
ROMANIA
CROATIA
Bosnia and Herzegovina
SERBIA
ITALY
Adriatic Sea
KOSOVO
BULGARIA
MACEDONIA (FYROM)
ALBANIA
Balearic Sea
SPAIN
MOUNT ETNA
Aegean Sea
Greece
Sicily
Ionian Sea
Alboran Sea

# CLASSIFICATION OF VOLCANOES

## 1 CLASSIFICATION ON THE BASIS OF PERIODICITY OF ERUPTION

### DORMANT VOLCANOES

Mt Vesuvius in Italy

1. Dormant Volcanoes are those which become quite after their eruptions for some time and there is no indications for future eruptions, but, suddenly they erupt very violently and cause enormous damage to human health.
2. Vesuvius Volcano is the best example of dormant volcano.
3. It erupted first in 79 AD and it had subsequent eruptions in 1631 AD, 1803, 1872, 1906, 1927,1928,1929 and 1944.


GERMANY
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CZECH REP
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Bosnia and Herzegovina
SERBIA
ITALY
Adriatic Sea
KOSOVO
BULGARIA
MACEDONIA (FYROM)
ALBANIA
Balearic Sea
MOUNT VESUVIUS
SPAIN
Aegean Sea
Greece
Ionian Sea
Alboran Sea

# CLASSIFICATION OF VOLCANOES

## 1 CLASSIFICATION ON THE BASIS OF PERIODICITY OF ERUPTION

### EXTINCT VOLCANOES


*Mt Kilimanjaro is believed to be extinct now*

1. The Volcanoes are considered extinct when there are no indications of future eruptions.
2. The crater is filled up with water and lakes are formed.
3. It may be pointed out that no volcano can be declared permanently dead as no one knows what is happening below its ground surface.
4. Mt Krakatau which was once thought by people to be extinct erupted most violently.


SUMATRA
INDONESIA
SUNDA STRAIT
Indian Ocean
Java Sea
MOUNT KRAKATAU
JAVA

# CLASSIFICATION OF VOLCANOES

## 2 CLASSIFICATION ON THE BASIS OF THE NATURE OF VOLCANIC ERUPTIONS

1

2

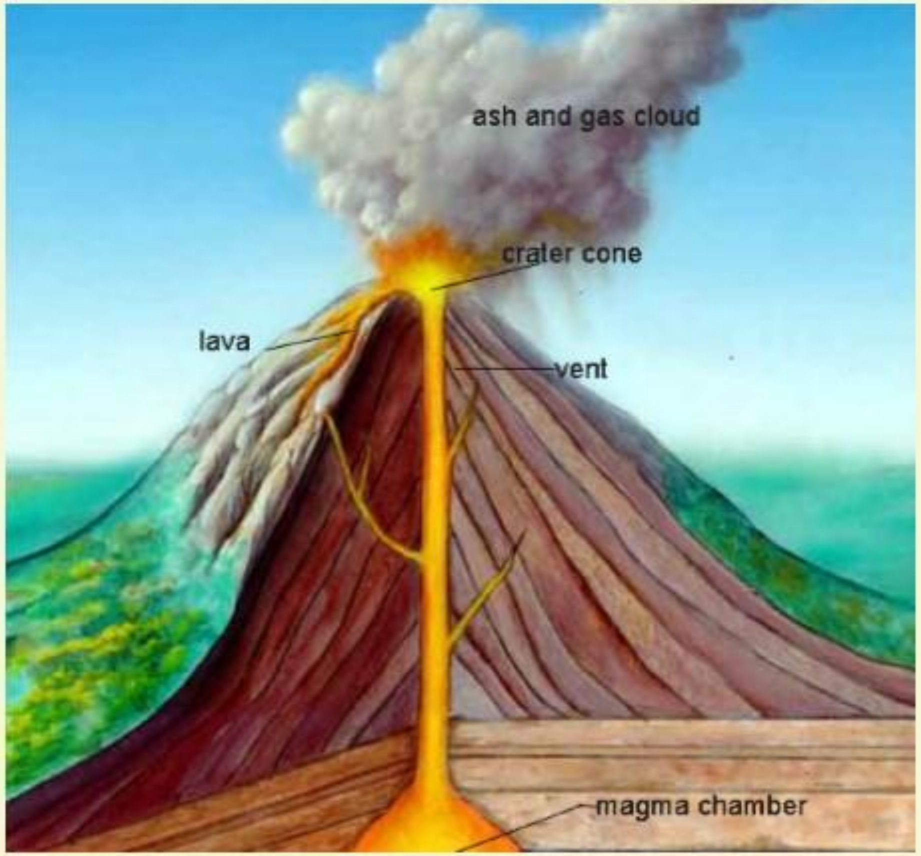
Volcanic eruptions occur mainly in two ways:

1. Violent and explosive type of eruption through a narrow pipe and small opening under the impact of violent gases. This is known as the Central Eruption type.
2. Eruption along a long fracture or fissure due to weak gases and huge volume of lavas. This is known as the Fissure type of eruption.

# CLASSIFICATION OF VOLCANOES

## 2 CLASSIFICATION ON THE BASIS OF THE NATURE OF VOLCANIC ERUPTIONS

### VOLCANOES OF CENTRAL ERUPTION TYPE

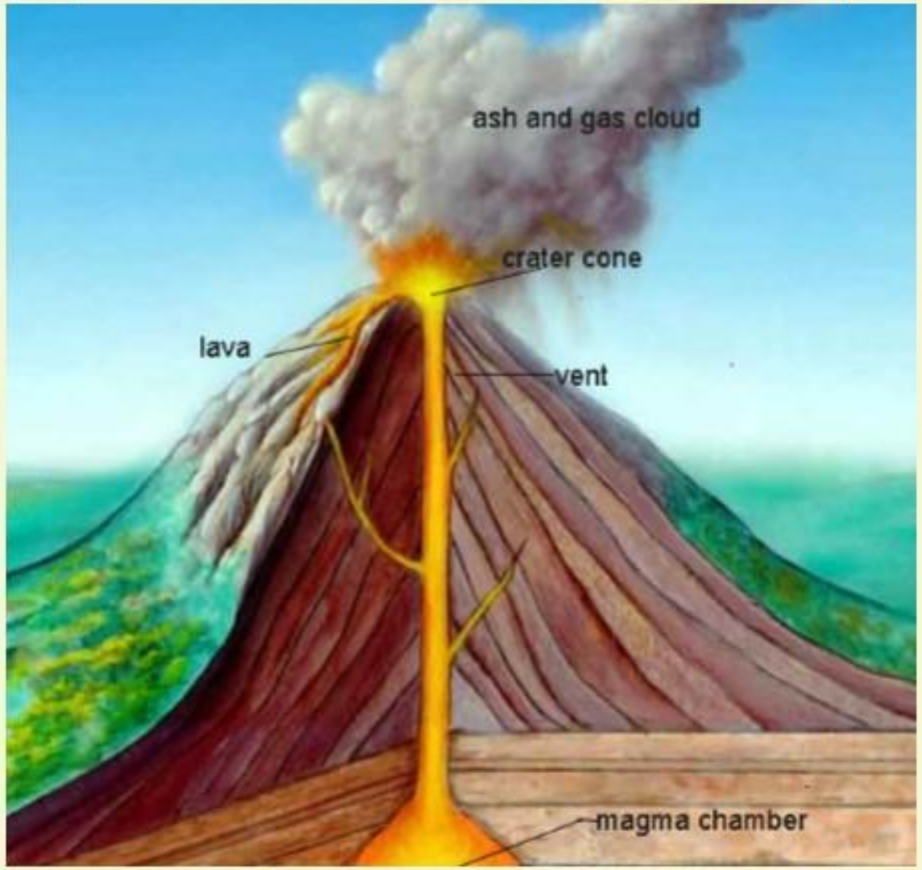



1. Central eruption type occurs through a central pipe and a small opening by breaking and blowing off crustal surface due to violent and explosive gases accumulated deep within the Earth.
2. Eruption is rapid and violent and fragmental materials are ejected up to 1,000 mts in the sky
3. The materials fall down and accumulate around the volcanic vent forming volcanic cones of various sizes.
4. Such Volcanoes are very destructive and cause disastrous natural hazards.

# CLASSIFICATION OF VOLCANOES

## 2 CLASSIFICATION ON THE BASIS OF THE NATURE OF VOLCANIC ERUPTIONS

### VOLCANOES OF CENTRAL ERUPTION TYPE




The Volcanoes of the Central Eruption types are sub-divided into five types:

1. Hawaiian Type Volcanoes
2. Strombolian Type of Volcanoes
3. Vulcanian Type of Volcanoes
4. Pelean Type of Volcanoes (These are the most violent and most explosive type of Volcanoes)
5. Vesuvius Type of Volcanoes

# CLASSIFICATION OF VOLCANOES

## 2 CLASSIFICATION ON THE BASIS OF THE NATURE OF VOLCANIC ERUPTIONS

### VOLCANOES OF CENTRAL ERUPTION TYPE

1. **Hawaiian type** - here extremely fluid lava comes out on the surface of the Earth. It is mostly basaltic and spread over large areas.

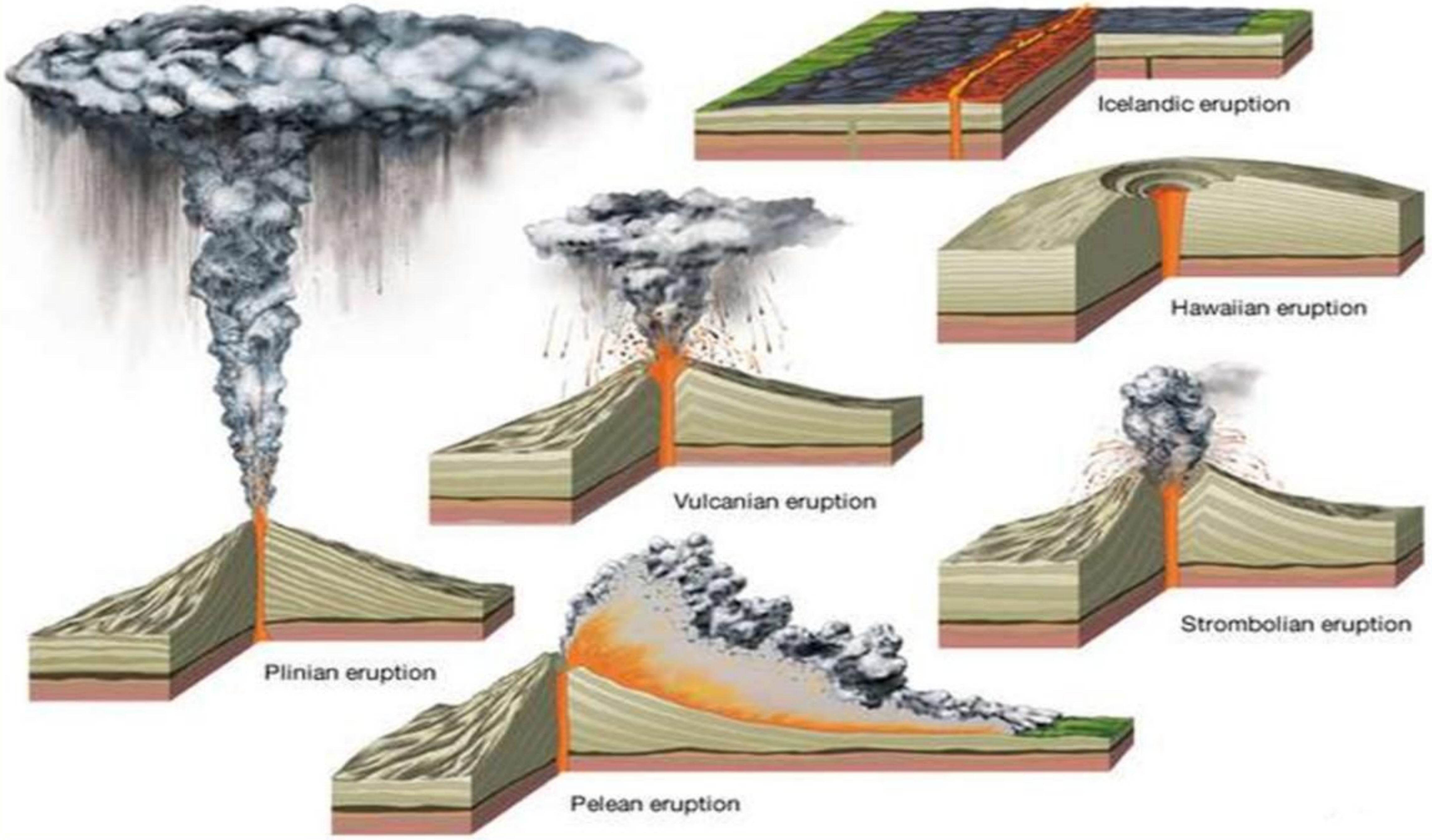
2. **Strombolian type** – they are the most typical volcano which are comparatively less explosive. There are several small explosive activities, where lava is more viscous. It has a power to explode the crust.

3. **Volcanian type** – here the lava is very viscous and the eruptions take place at longer interval. Large quantities of pyroclastic are erupted from them.

4. **Vesuvian type/Plinian type** –Here there is an extremely violent expulsion of magma due to enormous volume of explosive gases. Volcanic matter are thrown upto greater height in the sky forming 'cauliflower form' of clouds.

5. **Pelean type** - here the lava is the most viscous and highly destructive. Every successive eruption has to blow off these lava domes. Consequently, each successive eruption occurs with greater force and intensity.
   Eg : Mt Pelee on the Matinique island in West Indies

Space


Icelandic eruption
Hawaiian eruption
Vulcanian eruption
Strombolian eruption
Plinian eruption
Pelean eruption

# CLASSIFICATION OF VOLCANOES

## 2 CLASSIFICATION ON THE BASIS OF THE NATURE OF VOLCANIC ERUPTIONS

### VOLCANOES OF FISSURE ERUPTION TYPE

1. Such volcanoes occur along a long fracture/fissure and there is slow appearing of magma from below.
2. The resultant lavas spread over a large ground surface.
3. Speed of the lava movement depends on the nature of the magma, volume of magma, slope of the ground surface and temperature conditions.
4. Example: The Laki fissure eruption of 1783 in Iceland.

# CLASSIFICATION OF VOLCANOES

## 1 CLASSIFICATION ON THE BASIS OF PERIODICITY OF ERUPTION

Mt Etna in Sicily

### ACTIVE VOLCANOES

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